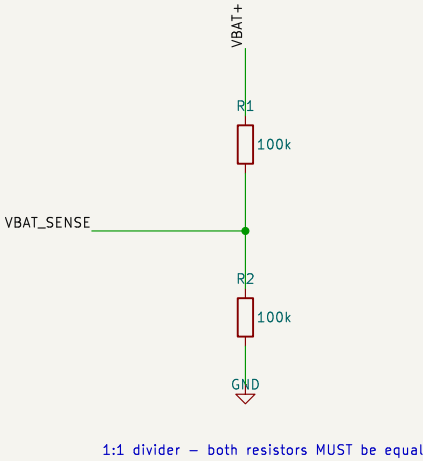
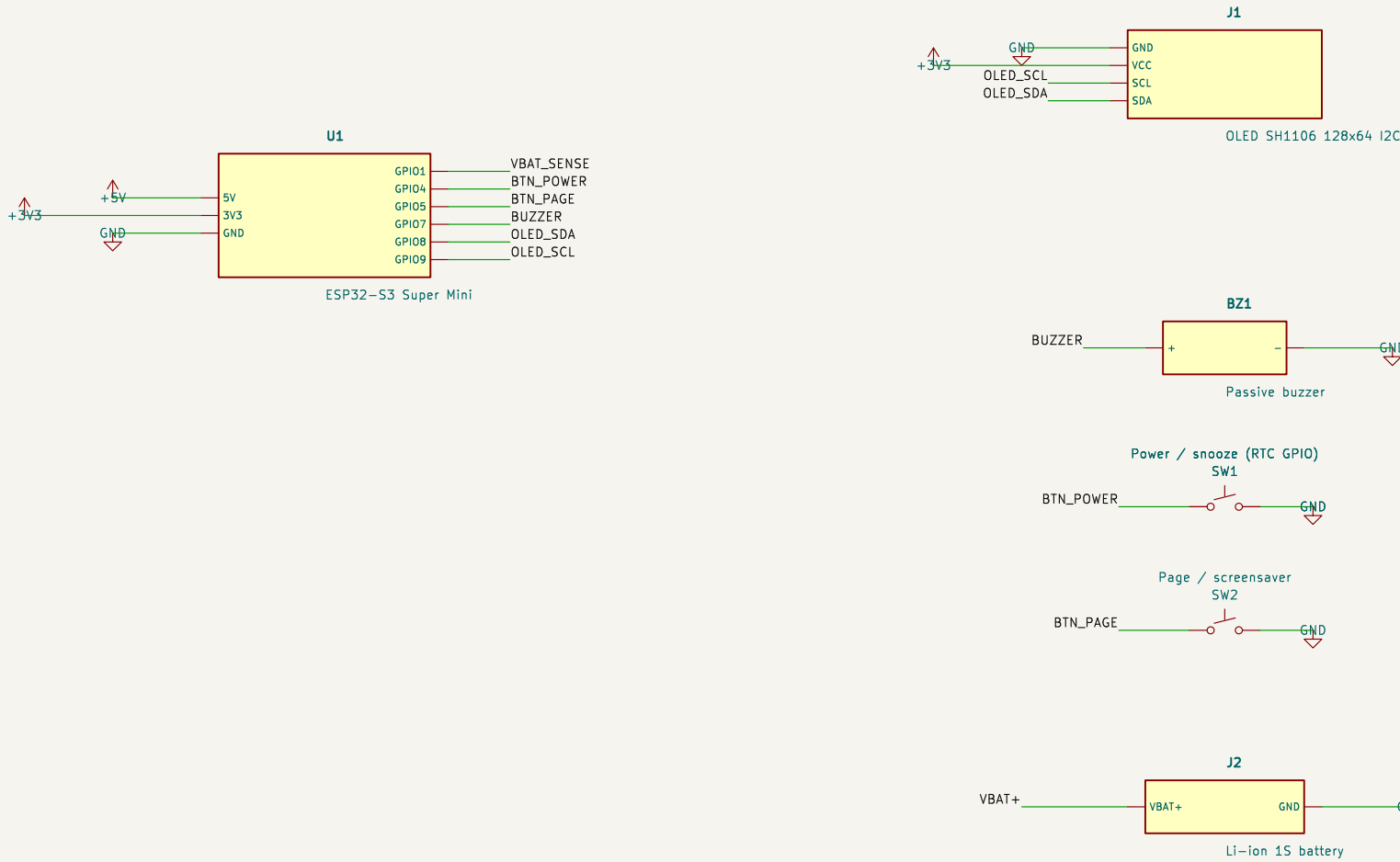




ASSEMBLY NOTES

- All GPIO numbers are read directly from firmware v4.7.5–OLED by the generator script; they are never typed in by hand. Re–run tools/kicad–gen/gen_schematics.py after a pin change.
- Both buttons use pinMode(INPUT_PULLUP), so they wire straight to GND. No external pull–ups.
- SW1 MUST sit on an RTC GPIO (0–21): it wakes the board from deep sleep via ext0 at logic level 0.
- Most OLED breakout modules already carry the I2C pull–ups. If yours does not, add 4.7k from SDA to 3V3 and 4.7k from SCL to 3V3.
- The I2C bus runs at 400 kHz per the SH1106 datasheet. Keep both wires as short as practical.
- R1 and R2 must be equal. The firmware computes (raw/4095) x 3.3 x 2.0, which fixes the ratio at 1:1.
- A full 4.2V cell divides down to 2.1V, comfortably inside the ESP32–S3 ADC input range.